Shravan Kumar

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Summary

Automotive diagnostics engineer with ∼4 years at TCS (currently onsite in Tokyo, Japan), building C++ diagnostic middleware and in-house ODX/OTX authoring tools. Hands-on with UDS, DoIP and CAN/CAN-FD across the ISO 22900 / 22901 / 13209 diagnostic standards, plus full-stack GenAI (RAG) delivery.

Skills

Languages: C++ (14/17/20), CAPL
Diagnostics & Standards: UDS (ISO 14229), DoIP (ISO 13400), CAN/CAN-FD (ISO 15765), OBD (SAE J1979), D-PDU (ISO 22900-2), D-Server (ISO 22900-3), ODX/PDX (ISO 22901), OTX (ISO 13209)
Frameworks & Tools: iceoryx, REST, GoogleTest, CMake, SonarQube, Git, AWS, Vector CANoe / vTESTstudio
Hardware: Raspberry Pi, Vector VN5620 / VN5610 / VN16xx, MVCI, TI Board

Experience

Tata Consultancy Services (TCS) — System Engineer Aug 2022 – Present | Tokyo, Japan

Vehicle Data Collector (VDC) — *Architect*

- Designed an onboard, config-driven data collector for a robotaxi (C++17, QNX-safe) — CAN and SOVD/UDS collection as live telemetry or bulk recordings, with offline store-and-forward.
- Defined the threading architecture — six stage workers, thread-per-bus CAN ingest, and bounded single-owner queues with explicit overflow policies.
- Specified verification across Windows/MSVC and Linux/GCC under ASan/UBSan; 1200+ tests green, with acceptance scored against the shipped binary.

Diagnostic Middleware (D-Server) — *Module Lead*

- Architected and led a C++17/20 D-Server middleware (ISO 22900-3) that runs ∼30% faster than the prior synchronous design — RAII ownership and `std::jthread` throughout, leak-free under Valgrind.
- Integrated CAN/CAN-FD and DoIP diagnostics over D-PDU (ISO 22900-2) channels, serving UDS and OBD-II through a factory/adapter layer that made new protocols drop-in.
- Enforced MISRA compliance in CI via SonarQube static analysis; led architecture reviews and mentored the team on modern, memory-safe C++.

Service-Oriented Vehicle Diagnostics (SOVD) — *C++ Developer*

- Built a modular C++ SOVD package on Raspberry Pi (4G/RTC/CAN HAT) for field-deployable remote diagnostics, with mDNS/DNS-SD discovery and SSL-based vehicle authentication.
- Delivered a containerized server with Swagger-documented REST APIs over secure AWS (Lambda, MQTT); used iceoryx zero-copy IPC for up to 5x faster inter-module messaging vs the prior design.

In-House Diagnostic Authoring Tools (ODX Studio & OTX) — *Developer*

- Modeled a diagnostics-oriented project (DIDs, DTCs, routines, sessions, security) with Excel/JSON import and one-click PDX export, as an open alternative to Vector CANdela Studio.
- Building a visual OTX authoring tool (Electron/React, ISO 13209) — a drag-and-drop flowchart canvas generating standards-compliant OTX, targeting full Part 2 + Part 3 library coverage.

OneConnect — AI Vehicle Diagnostic Assistant — *Full-Stack Developer*

- Built a GenAI diagnostic assistant: a Blazor/.NET front end with a Python FastAPI RAG backend over AWS Bedrock (Claude), OpenSearch, DynamoDB, S3, and Polly.
- Grounded responses in live vehicle data via the SOVD protocol for real-time ECU/DTC insight.

ECU Simulation & Validation Toolchain — *Developer*

- Validated DTCs, DIDs, RIDs and diagnostic routines, tracing faults across ECUs over UDSonCAN and UDSonDoIP (Vector CANoe, vTESTstudio).
- Configured 50+ simulated ECUs and benchmarked gateways over OBDonUDS (SAE J1979); used Vector hardware (VN5620/VN16xx) for repeatable, vehicle-level validation.

Education

B.Tech, Electronics & Telecommunication Engineering — Bharati Vidyapeeth College of Engineering
Jul 2018 – Jul 2022 | GPA 8.92/10

Japanese-Language Proficiency Test (JLPT) N5 Jul 2026

Achievements

- Best Performer (twice) at TCS for innovation in diagnostics tooling and middleware; recognized for process improvement toward zero-defect delivery. Deployed onsite to Nissan, Tokyo.